Velocity gradient prediction using parameterized Lagrangian deformation models

Criston Hyett [1] Yifeng Tian [3] Misha Stepanov [1] Daniel Livescu [2]
Misha Chertkov[1]

[1]Program in Applied Mathematics, University of Arizona

[2]CCS-2, Los Alamos National Labs

[3]CCS-3, Los Alamos National Labs

November 19, 2023

Summary

- Velocity Gradient Tensor
- Deformation Models
- Baseline Model: Tensor Basis Neural Network
- Extending the TBNN: Parameterized Deformation Models
- Results

Velocity Gradient Tensor

- The **velocity gradient tensor** (VGT) is defined by $A_{ij} = \frac{\partial u_i}{\partial x_j}$, so from Navier Stokes

$$\frac{\partial u_i}{\partial t} + u_k \frac{\partial u_i}{\partial x_k} = -\frac{\partial P}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_k \partial x_k} \tag{1}$$

- We apply spatial derivatives, and use incompressibility to find the governing equations for the **Lagrangian VGT**:

$$\frac{dA_{ij}}{dt} = -\boxed{\left(A_{ik}A_{kj} - \frac{1}{3}A_{mn}A_{nm}\delta_{ij}\right)} - \boxed{\left(\frac{\partial^2 P}{\partial x_i \partial x_j} - \frac{1}{3}\frac{\partial^2 P}{\partial x_k \partial x_k}\delta_{ij}\right)} + \boxed{\nu \frac{\partial^2 A_{ij}}{\partial x_k \partial x_k}} \tag{2}$$

- The main challenge is to predict the nonlocal **deviatoric pressure hessian** (PH), defined as

$$H_{ij} = -\boxed{\left(\frac{\partial^2 P}{\partial x_i \partial x_j} - \frac{1}{3}\frac{\partial P}{\partial x_k \partial x_k}\delta_{ij}\right)} \tag{3}$$

Lagrangian Deformation Models

Phenomenological ideas have tried to relate the **deformation tensor** to pressure Hessian predictions

$$\frac{dD_{ij}}{dt} = A_{ik} D_{kj}, \qquad \text{with } D_{ij}(0) = \delta_{ij}, \tag{4}$$

- Chertkov et.al. used deformation in the **Tetrad Model** to resolve the finite time singularity induced by the Restricted Euler approximation to the pressure Hessian (1999)

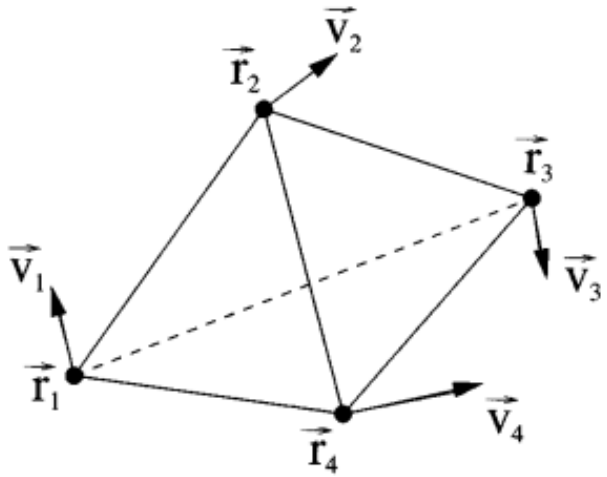


Figure: Taken from Chertkov et.al. 1999 *Lagrangian tetrad dynamics and the phenomenology of turbulence*

Lagrangian Deformation Models

Phenomenological ideas have tried to relate the **deformation tensor** to pressure Hessian predictions

$$\frac{dD_{ij}}{dt} = A_{ik} D_{kj}, \qquad \text{with } D_{ij}(0) = \delta_{ij}, \tag{4}$$

- Chevillard & Meneveau, used the deformation tensor and a short-time hypothesis to create the **Recent Fluid Deformation** model (2006)
- Johnson & Meneveau enriched RFD with a better upstream condition, resulting in **Recent Deformation of Gaussian Fields** closure (2016)

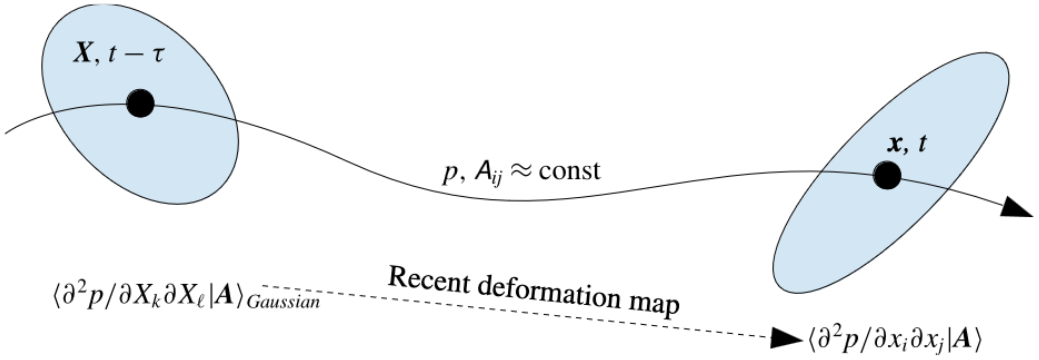


Figure: Taken from Johnson & Meneveau 2016 *A closure for Lagrangian velocity gradient evolution in turbulence using recent-deformation mapping of initially Gaussian fields*

Lagrangian Deformation Models

Phenomenological ideas have tried to relate the **deformation tensor** to pressure Hessian predictions

$$\frac{dD_{ij}}{dt} = A_{ik} D_{kj}, \quad \text{with } D_{ij}(0) = \delta_{ij}, \tag{4}$$

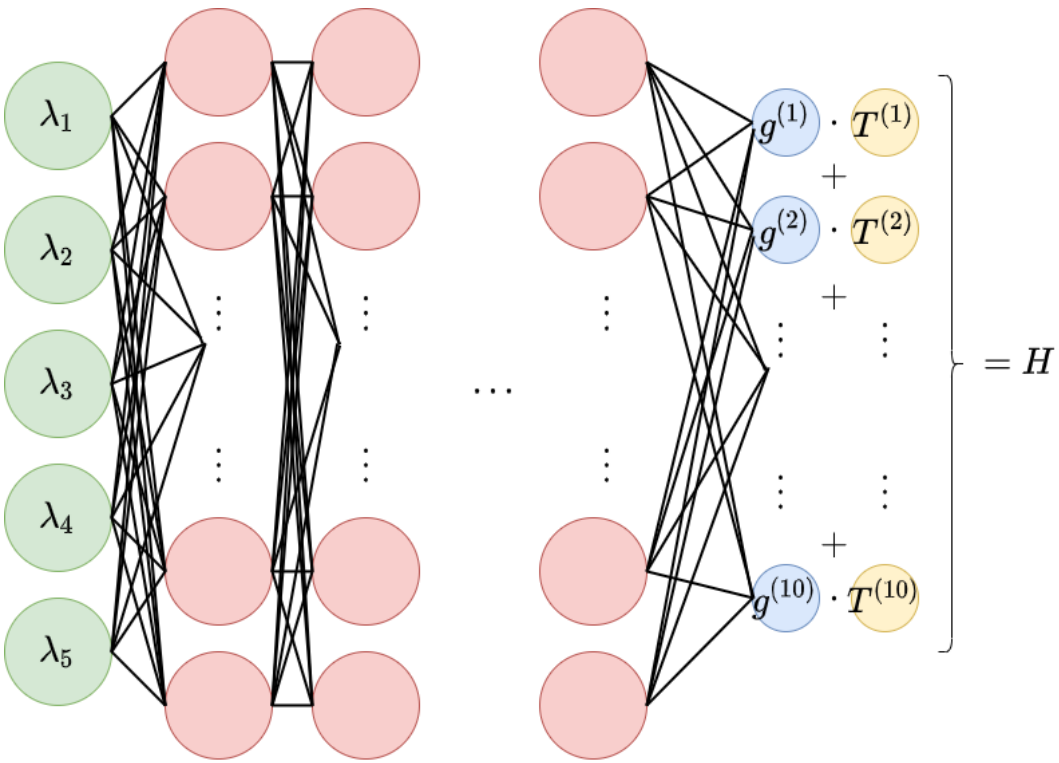
- Chertkov et.al. used deformation in the **Tetrad Model** to resolve the finite time singularity induced by the Restricted Euler approximation to the pressure Hessian (1999)
- Chevillard & Meneveau, used the deformation tensor and a short-time hypothesis to create the **Recent Fluid Deformation** model (2006)
- Johnson & Meneveau enriched RFD with a better upstream condition, resulting in **Recent Deformation of Gaussian Fields** closure (2016)

- Can we **learn** what history is important?

Baseline model: Tensor Basis Neural Network

Tian et.al. used physics-informed machine learning to learn a local (in time and space) closure to the PH

$$\hat{H} = \sum_{i=1}^{10} g_{\theta}{}^{(i)}(\lambda_1, \ldots, \lambda_5) \cdot \hat{T}^{(i)}(\hat{A}) \tag{5}$$



Parameterized Deformation Model

- Taking the deformation tensor as inspiration, we introduce a new object,

$$X = \sum_{j=-N}^{0} \alpha_j A(t_j) \tag{6}$$

  with α_j learnable parameters, and $\Delta t \approx \tau_{Kolmogorov}$
- We extend the TBNN using historical information via:

$$g_\theta{}^{(i)}(\lambda_1, \ldots, \lambda_5) \rightarrow g_\theta{}^{(i)}(\lambda_1, \ldots, \lambda_5, X) \tag{7}$$

- Train and test data is **incompressible, isotropic turbulence** at $Re \approx 240$

Qualitative Results: *QR* CMT

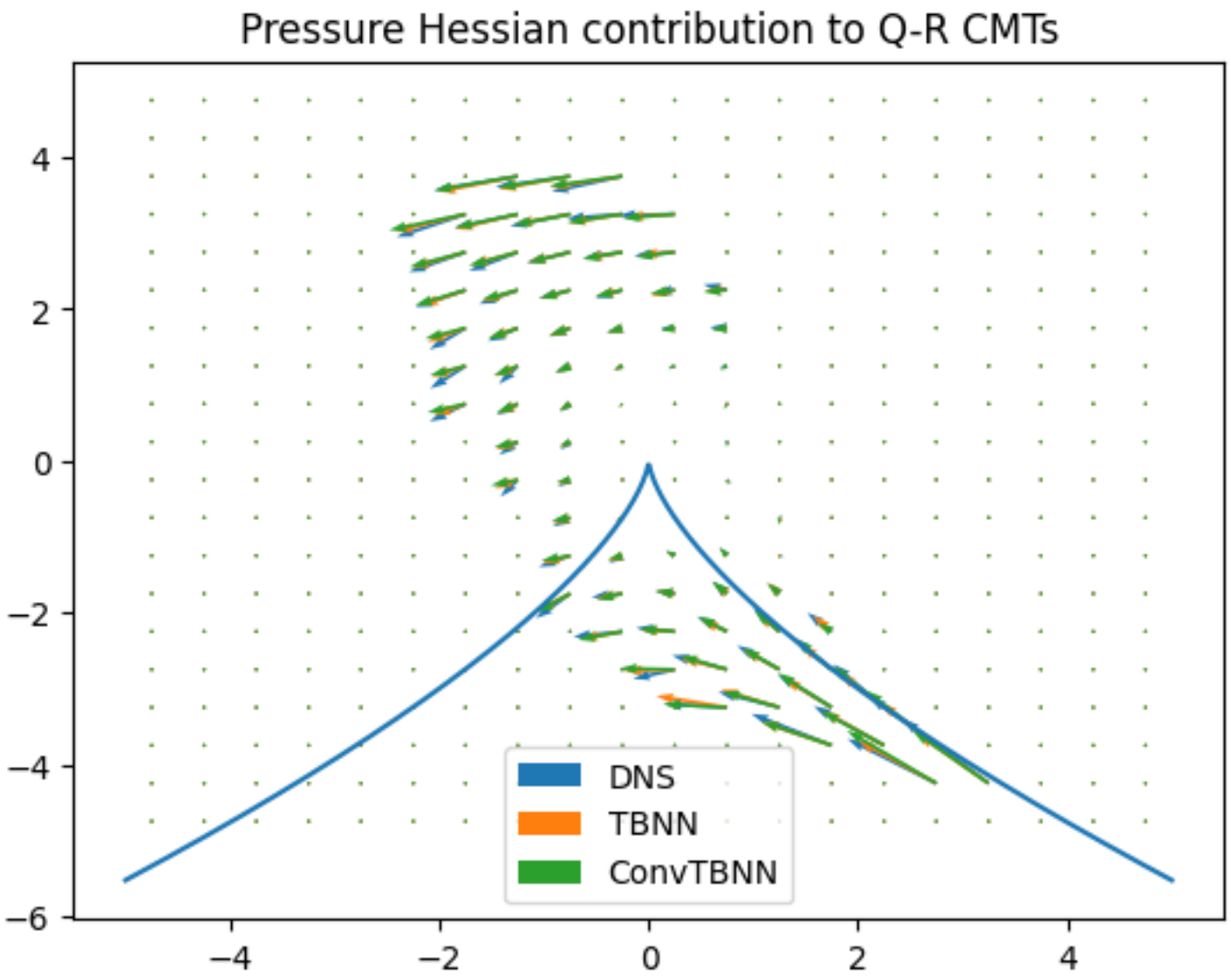


Figure: Qualitatively, the *QR* conditional mean tangents match the TBNN

Quantitative Results: Pressure Hessian Eigenvector Alignment

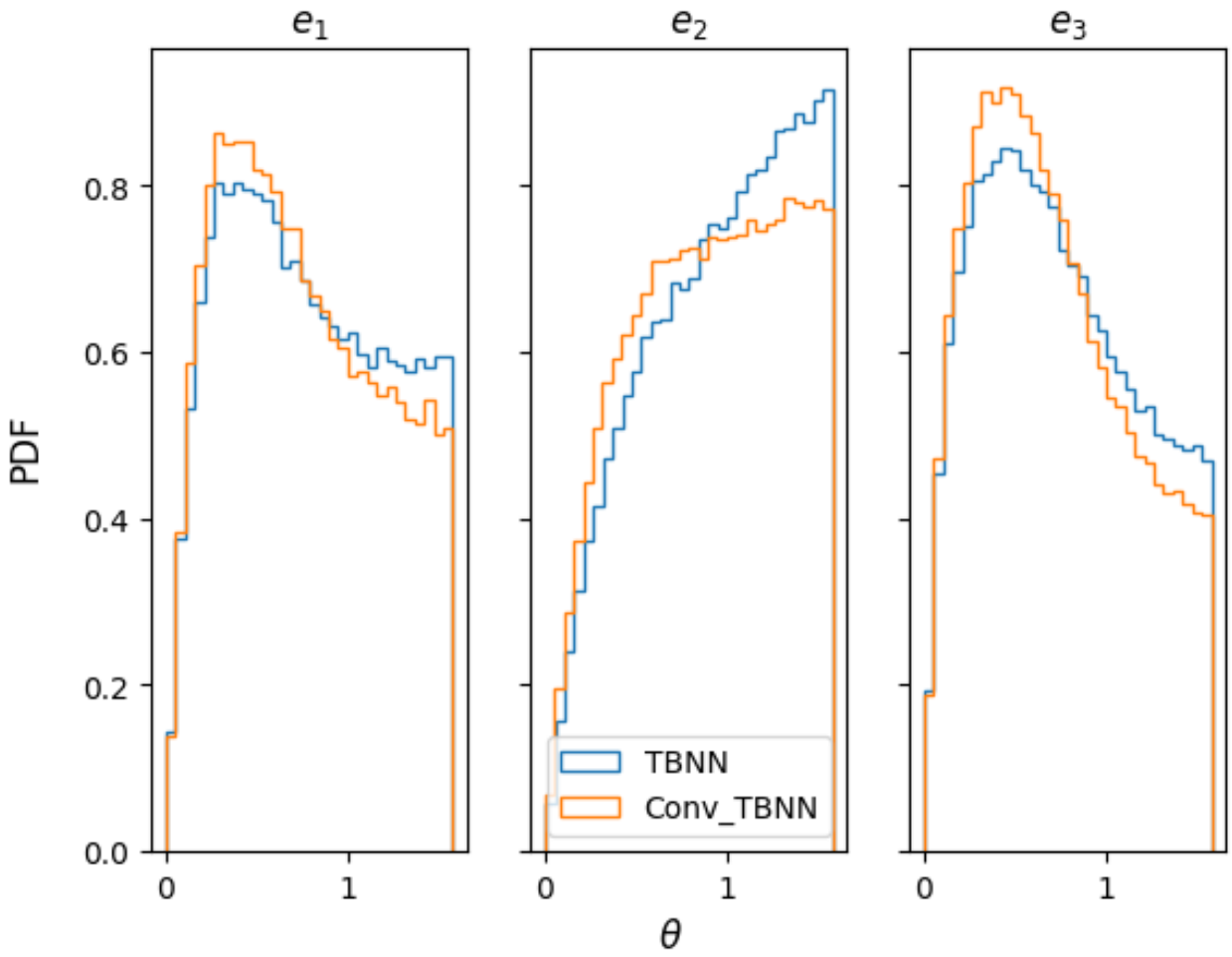
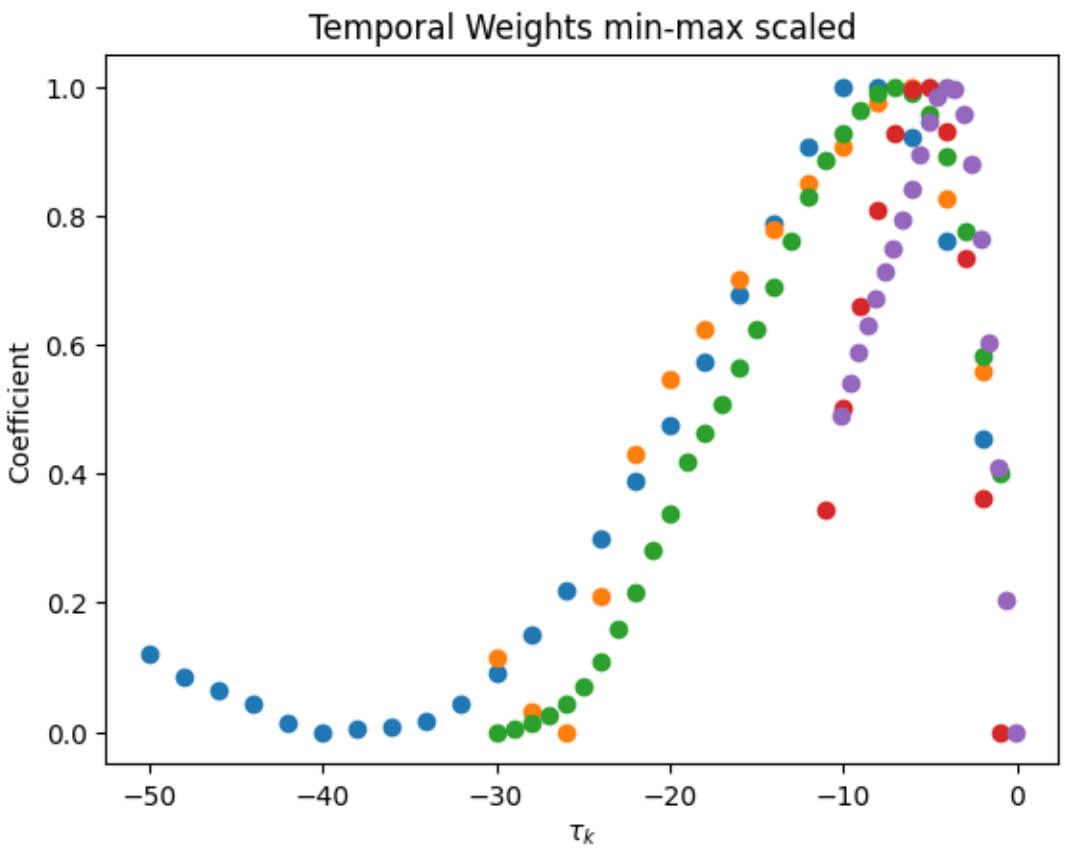


Figure: The ConvTBNN outperforms the TBNN in eigenvector alignment

Quantitative Results: Convolution kernel shape

$$X = \sum_{j=-N}^{0} \alpha_j A(t_j)$$



1. Weights are small in far history (and for short kernels no flat tails are observed)
2. Weights are small for very recent history (redundant?)
3. Weights in some importance region increase monotonically - reinforcing phenomenological ideas

Conclusion

- By utilizing the **history of the VGT**, we can **improve the prediction of the pressure Hessian**
- Trained convolution kernels suggest a **finite, but significant length of important history**
- Can we leverage learned kernels to inform theory?

Parameterized Deformation Model

$$\frac{dD_{ij}}{dt} = A_{ik} D_{kj}, \qquad \text{with } D_{ij}(0) = \delta_{ij}, \tag{8}$$

which has the formal solution

$$D(t) = \mathrm{Texp}\left(\int_0^t dt' A(t')\right) = \lim_{N\to\infty} \prod_{i=0}^{N} \left(e^{A(t_i)\Delta t}\right), \tag{9}$$

If we only take the leading order term of the discrete form:

$$D(t) \approx \left(I + \Delta t \cdot \sum_{i=0}^{N} A(t_i)\right) \tag{10}$$

Taking this as motivation, we introduce a new object,

$$X = \sum_{j=-N}^{0} \alpha_j A(t_j) \tag{11}$$

and the α_j are learnable parameters, referred to as **convolution kernel** with $\Delta t \approx \tau_{Kolmogorov}$

Parameterized Deformation Model

$$X = \sum_{j=-N}^{0} \alpha_j A(t_j) \tag{8}$$

- We extend the TBNN using historical information via:

$$g_\theta{}^{(i)}(\lambda_1, \ldots, \lambda_5) \to g_\theta{}^{(i)}(\lambda_1, \ldots, \lambda_5, X) \tag{9}$$